# Molecular reaction dynamics

**Molecular reaction dynamics** is the study of **chemical and physical transformations of matter at the molecular level**.

It involves understanding
*how chemical reactions occur
*and how to control them,

which is fundamental to chemists and interdisciplinary areas such as materials science, nanoscience, drug design, and environmental chemistry*

## Reactive encounters

We consider two elementary approaches to the calculation of reaction rates,

**one relating to gas-phase reactions**
**and the other to reactions in solution.**

Both approaches are based on the view that reactant molecules must meet, and that reaction takes place only if the molecules have a certain minimum energy.

# Collision theory

Purpose: Calculation of rate constants for simple elementary reactions.

For reactions to take place:

1. Reactant molecules must meet.
2. Must hold a minimum energy.

Gas phase reactions: Collision theory.

Solution phase reactions: Diffusion controlled.

Activation controlled.

## Collision theory

We shall consider the bimolecular elementary reaction

$$\mathbf{A + B \rightarrow P} \qquad\qquad \boldsymbol{v = k_2[A][B]}$$

where P denotes products, and aim to calculate the second-order rate constant $k_2$.

We can anticipate the general form of the expression for $k_2$ by considering the physical requirements for reaction. We expect the rate $v$ to be proportional to the rate of collisions, and therefore to the mean speed of the molecules, $\boldsymbol{K \propto (T/M)^{1/2}}$

Where $M$ is the molar mass of the molecules, their collision cross-section, σ, and the number densities $N_A$ and $N_B$ of A and B:

$$\nu \propto \sigma(T/M)^{1/2} N_A N_B \propto \sigma\,(T/M)^{1/2}[A][B]$$

However, a collision will be successful only if the kinetic energy exceeds a minimum value, the activation energy, *E*a, of the reaction. This requirement suggests that the rate constant should also be proportional to a Boltzmann factor of the form $e^{-Ea/RT}$.

So we can anticipate, by writing the reaction rate in the form given , that

$$k_2 \propto \sigma\,(T/M)^{1/2}\, e^{-Ea/RT}$$

Not every collision will lead to reaction

Not every collision will lead to reaction even if the energy requirement is satisfied, because the reactants may need to collide in a certain relative orientation.

This ‘steric requirement’ suggests that a further factor, *P*, should be introduced, and that

$$k_2 \propto P\sigma\,(T/M)^{1/2}\mathrm{e}^{-Ea/RT}$$

As we shall see in detail below, this expression has the form predicted by collision theory.

It reflects three aspects of a successful collision:

**$k_2$ ∝ steric requirement × encounter rate × minimum energy requirement**

# Collision rate in gases

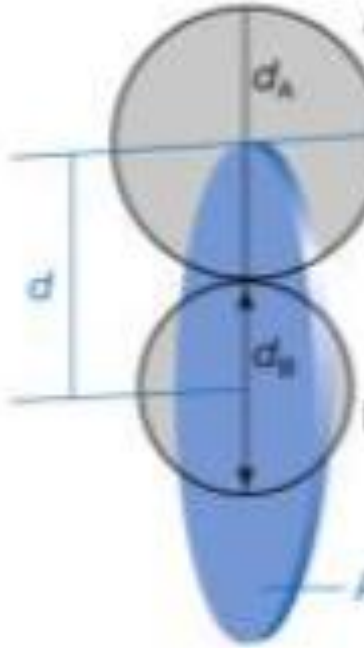
- ***Collision density***, $Z_{AB}$, is the number of (A, B) collisions in a region sample in an interval of time divided by the volume of the region and duration of the interval.

$$Z_{AB} = \sigma\left(\frac{8kT}{\pi u}\right)^{1/2} N_A^2[A][B]$$

where $\sigma = \pi d^2$ $\quad d = \frac{1}{2}(d_A + d_B)$

and $u$ is the reduced mass

$$u = \frac{m_A m_B}{m_A + m_B}$$




- when A and B are the same, one gets

$$Z_{AA} = (1/2)\sigma\left(\frac{4kT}{\pi u}\right)^{1/2} N^2[A]^2$$

# Diffusion-controlled reactions

Encounters between reactants in solution occur in a very different manner from encounters in gases.

Reactant molecules have to jostle their way through the solvent, so their encounter frequency is considerably less than in a gas.

However, because a molecule also migrates only slowly away from a location, two reactant molecules that encounter each other stay near each other for much longer than in a gas.

This lingering of one molecule near another on account of the hindering presence of solvent molecules is called the **cage effect**.

Such an encounter pair may accumulate enough energy to react even though it does not have enough energy to do so when it first forms.

The activation energy of a reaction is a much more complicated quantity in solution than in a gas because the encounter pair is surrounded by solvent and we need to consider the energy of the entire local assembly of reactant and solvent molecules.

## Classes of reaction

The complicated overall process can be divided into simpler parts by setting up a simple kinetic scheme. We suppose that the rate of formation of an encounter pair AB is
first-order in each of the reactants A and B:

$A + B \rightarrow AB \qquad v = k_d[A][B]$

As we shall see, $k_d$ is determined by the diffusional characteristics of A and B. The encounter pair can break up without reaction or it can go on to form products P. If we suppose that both processes are pseudo first-order reactions (with the solvent perhaps playing a role), then we can write

$AB \rightarrow A + B \qquad v = k_d'[AB]$

and

$AB \rightarrow P \qquad v = k_a[AB]$

The concentration of AB can now be found from the equation for the net rate of change of concentration of AB:

$$= k_d[A][B] - k_d'[AB] - k_a[AB] \approx 0$$

where we have applied the steady-state approximation. This expression solves to

$$[\mathrm{AB}] = \frac{k_\mathrm{d}[\mathrm{A}][\mathrm{B}]}{k_\mathrm{a} + k'_\mathrm{d}} \tag{24.22}$$

The rate of formation of products is therefore

$$\frac{\mathrm{d}[\mathrm{P}]}{\mathrm{d}t} \approx k_\mathrm{a}[\mathrm{AB}] = k_2[\mathrm{A}][\mathrm{B}] \qquad k_2 = \frac{k_\mathrm{a}k_\mathrm{d}}{k_\mathrm{a} + k'_\mathrm{d}} \tag{24.23}$$

Two limits can now be distinguished. If the rate of separation of the unreacted encounter pair is much slower than the rate at which it forms products, then $k'_\mathrm{d} \ll k_\mathrm{a}$ and the effective rate constant is

$$k_2 \approx \frac{k_\mathrm{a}k_\mathrm{d}}{k'_\mathrm{d}} = k_\mathrm{a}K$$

## (b) Diffusion and reaction

The rate of a diffusion-controlled reaction is calculated by considering the rate at which the reactants diffuse together.

The rate constant for a reaction in which the two reactant molecules react if they come within a distance $R^*$ of one another is

$$k\mathrm{d} = 4\pi R^* DN\mathrm{A}$$

Where $D$ is the sum of the diffusion coefficients the two reactant species in the solution.